Labor Input Responses to Ownership Changes: Evidence from U.S. Nursing Homes

Joseph White

Miami University

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Motivation: Ownership Changes Beyond Market Structure

- Ownership transitions are common in healthcare.
- IO research often emphasizes **prices, market power, and concentration**.
- But many transitions do **not** change the set of providers in a market.
- In those cases, effects operate through **internal production decisions**: management practices and especially **staffing inputs**.

Research Question

- **Research question:** How do nurse staffing levels respond to ownership changes in nursing homes?
  - Do responses differ across staff type or organizational structure?
  - Do responses differ pre vs post-pandemic?
- Why we care: staffing is the primary input into care, so ownership changes can affect **quality** even when market structure is unchanged.

Why Nursing Homes?

- Nursing home care is labor-intensive and heavily regulated.
- **Nursing labor is the main input** into care and a major operating cost.
- Staffing is a natural adjustment margin after ownership change, but it's constrained by:
  - compliance,
  - resident outcomes,
  - demand/occupancy

Nursing Staff Types

- We study four staffing measures:
  - **RN (Registered Nurses):** clinical assessment, care planning, meds, supervision
  - **LPN (Licensed Practical Nurses):** routine clinical tasks, meds under RN oversight
  - **CNA (Certified Nursing Assistants):** direct daily care
  - **Total nursing:** RN + LPN + CNA
- Ownership change can affect **levels** (total hours) and **mix** (skill composition).

The Cost–Quality Tradeoff in Staffing

- Owners can raise quality through:
  - **Quantity** of staffing and/or
  - **Quality / skill mix**
- Either margin increases costs
- But **revenues are relatively flat in the short run**
- So owners face a tradeoff: **higher quality vs. tighter margins**.

Why Staffing Could Rise or Fall After a Transition

- **Cost pressures:** tighter budgets / debt service can push cost cutting.
- **Reorganization:** standardized policies may shift RN/LPN/CNA composition.
- **Quality objective:** owners with stronger quality/compliance goals may increase staffing.
- Net effect is ambiguous $\Rightarrow$ **empirical question**.

Data Sources and Unit of Observation

- **Healthcare Cost Report Information System (HCRIS)**
  - Annual cost report information
  - Contains information about costs, revenue, financial data and dates of ownership change
- **Nursing Home Compare (NHC) Archive**
  - Contains ownership information as well as information on chain affiliation
- **Payroll Based Journal (PBJ)**
  - Payroll-reported staffing hours by staff type
  - Used to construct outcome variables
- **Facility-month panel, 48 contiguous states, Jan 2017 – Jun 2024**

Identifying Ownership Changes: Step 1

- **Starting point (HCRIS):** includes a variable indicating **whether and when** an ownership change occurred.
- **Challenge:** ownership is often **layered** (organizations owning organizations), so an administrative flag may not map cleanly to an **economic change in control**.
- **Check:** NHC archive provides owner identities and structure:
  - **Owner name**, **ownership level**, and **owner type**

Ownership Level Challenge

- We identify ownership changes at the **highest level of ownership present**

Ownership Level	**Type**	**Owner Name**	**Share**
Direct ownership			
Organization	LLC	BREDLEGS HOLDINGS LLC	25%
Organization	LLC	ROCK HILL ACQUISITION III LLC	75%
Indirect ownership			
Individual	Person	LALLY, THOMAS	25%
Individual	Person	LITT, ALAN	25%
Individual	Person	LITT, JONATHAN	25%
Individual	Person	STURM, JOSHUA	25%

Identifying Ownership Changes: Step 2

- **Validate & date using NHC:**
  - Track ownership shares over time within the highest reported ownership level
  - Define a change when **majority control (50%)** transfers between $t-1$ and t
  - We do not flag changes driven by name changes or within-family reclassification
- **Example from the data:**
  - *Before:*
    - ESTES JAMES (84%)
    - BURCHFIELD JOHN (10%)
    - LEE CLAUDE (6%)
  - *After:*
    - JAMES N ESTES JR FAMILY DYNASTY TR NO 2 (50%)
    - JENNIFER E AGEE FAMILY DYNASTY TR NO 2 (50%)
  - We do **not** treat this as a meaningful ownership change

Dependent Variables: Nurse Staffing (PBJ)

- We study four staffing measures:
  - **Registered Nurses (RN)**: clinical assessment, medication administration, supervision
  - **Licensed Practical Nurses (LPN)**: clinical support, medication-related tasks
  - **Certified Nursing Assistants (CNA)**: direct daily care (bathing, feeding, mobility)
  - **Total nursing staff**: RN + LPN + CNA
- Outcome is **hours per patient day (HPPD)**: staffing intensity standardized by resident census

$$\text{HPPD}_{s,i,t} = \frac{H_{s,i,t}}{\text{PatientDays}_{i,t}}$$

Final Sample Construction

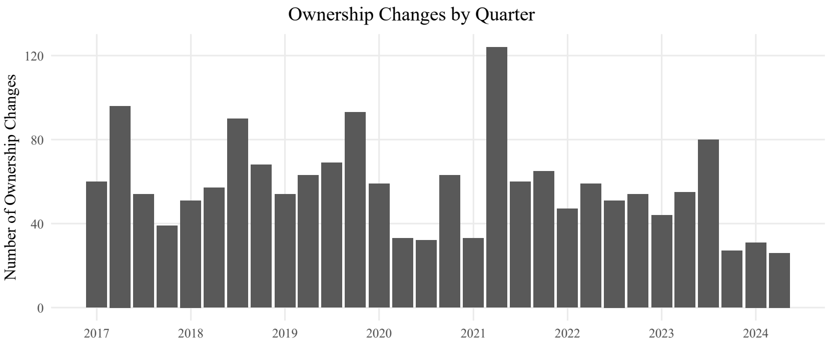
- **Validated ownership-change sample (HCRIS $\leftrightarrow$ NHC):**
  - Keep facilities with **(0,0)** changes in both datasets, or **(1,1)** with dates within 6 months
  - Event month is the **first month** NHC indicates a change
- **CMS implausible staffing**
  - RN HPPD $= 0$ **and** LPN HPPD $= 0$
  - Total HPPD < 1.5 or > 12
  - CNA HPPD > 5.25
- **Additional restrictions:** beds < 15; hospital-based facilities
- **Final sample:** 7,806 facilities; 632,231 facility-months; 1,589 are treated, about 2% of the final sample

Ownership Change Agreement: NHC vs HCRIS

Cross-tab of Ownership Changes

Count of changes in NHC \ Count of changes in HCRIS	0	1	2+
2+	1,137 (8.5%)	1,377 (10.3%)	535 (4.0%)
1	1,874 (14.0%)	1,589 (11.9%)	101 (0.8%)
0	6,217 (46.6%)	499 (3.7%)	15 (0.1%)

Ownership Changes Over Time (Quarterly)



Empirical Strategy: Event-Study Difference-in-Differences

- **Design:** staggered-adoption DiD around ownership transitions
- **Variation:** within-facility staffing changes relative to facilities *not yet treated*
- **Event study (dynamic effects):**

$$Y_{it} = \sum_{\tau=-24}^{24} \beta_\tau \, \mathbf{1}\{event\_time_{it} = \tau\} \cdot Treat_i + X'_{it}\gamma + \alpha_i + \lambda_t + \varepsilon_{it}.$$

- **Key objects:**
  - Y_{it}: staffing outcome in **HPPD** (RN, LPN, CNA, Total)
  - β_τ: effect at event time τ (relative to the reference period)
  - α_i facility FE; λ_t month FE; X_{it} controls (beds, occupancy, payer mix, ownership type, chain, acuity)
  - SEs clustered at the facility level

Empirical Strategy: TWFE Difference-in-Differences

- **Goal:** summarize the **average** staffing change after an ownership transition
- **TWFE DiD (headline effect):**

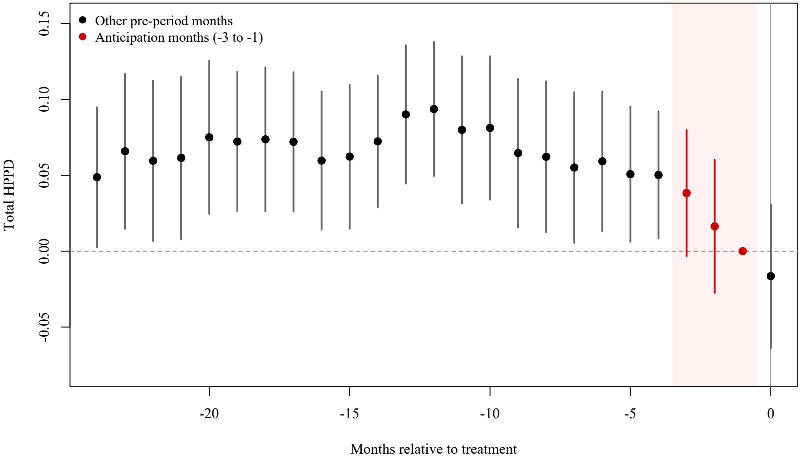
$$Y_{it} = \beta \, Post_{it} + X_{it}'\gamma + \alpha_i + \lambda_t + \varepsilon_{it}$$

- **Key objects:**
  - $Post_{it} = 1$ in all months after a change ($= 0$ before)
  - β: **average post-change effect** on staffing (HPPD)
  - Same fixed effects and controls as the event-study specification
  - SEs clustered at the facility level

Identification: Parallel Trends

- **Parallel trends:** absent an ownership change, treated facilities would have followed the same staffing trends as facilities not yet treated.
- **Why anticipation is plausible in ownership transitions**
  - Changes are not instantaneous: negotiations and operational handoffs precede the recorded month
  - Staffing can adjust during planning (turnover, hiring freezes, renegotiated contracts)
  - Recorded dates may reflect administrative/legal timing, not first operational change
- **Implication:** months just before $\tau = 0$ may violate parallel trends.

Evidence of violation in the Pre-Period



Main Specification: Donut Event Study + Pre-trend Test

- **Donut event study:** mitigate this influence by excluding months just before the recorded transition.
- **Implementation**
  - Exclude $\tau \in \{-3, -2, -1\}$
  - Use $\tau = -4$ as the reference period
- **Parallel Trends:** joint Wald test of pre-period leads

$$H_0 : \ \beta_\tau = 0 \quad \forall\, \tau < 0$$

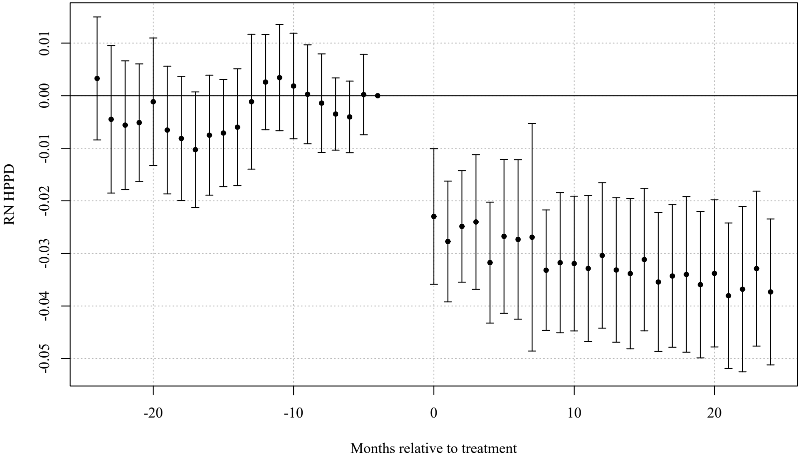
- **Takeaway:** full pre-window shows evidence of pre-trends; the donut specification is more consistent with parallel trends.

Parallel Trends Tests (Joint Wald) — Total Staffing

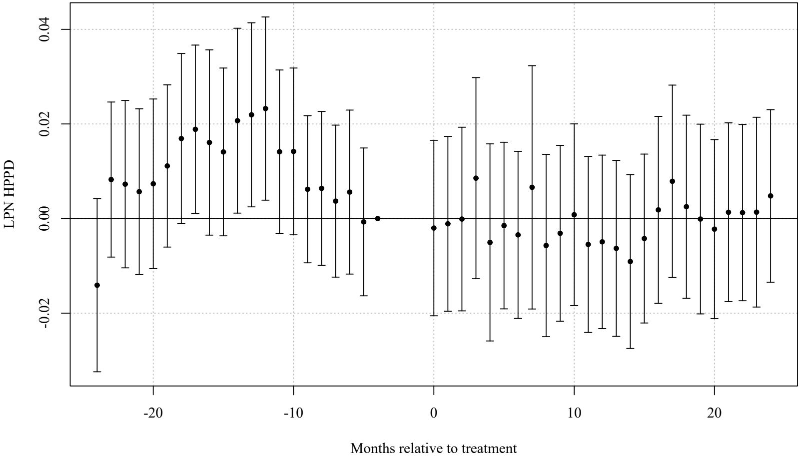
	Total (Levels)	**Total (Logs)**
2 Year Full Pre-Window	51.22 (23) [0.0006]	49.22 (23) [0.0012]
2 Year Window with Donut	24.92 (20) [0.2046]	22.26 (20) [0.3268]
1 Year Window with Donut	6.08 (8) [0.6387]	6.33 (8) [0.6102]

Notes: Joint Wald χ^2 test of $H_0 : \beta_\tau = 0$ for all pre-treatment coefficients. Cells report χ^2 (df) [p-value].
Donut excludes $\tau \in \{-3, -2, -1\}$; reference period is $\tau = -4$.
Full outcome-by-staff-type table available in appendix.

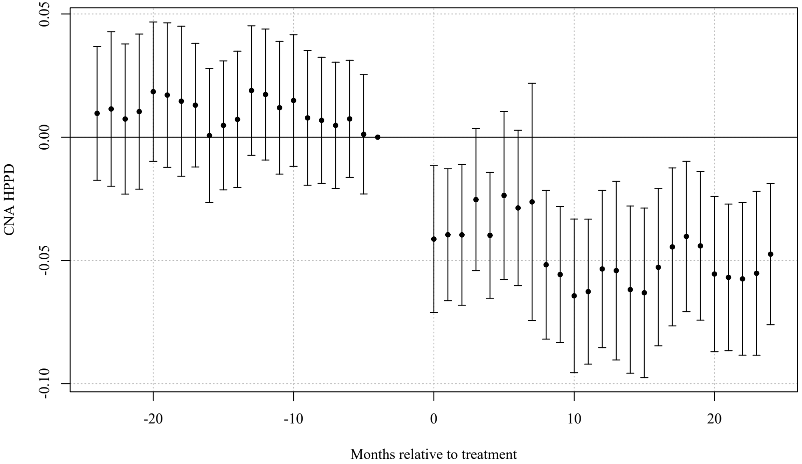
Main Results: Event Study (RN HPPD)



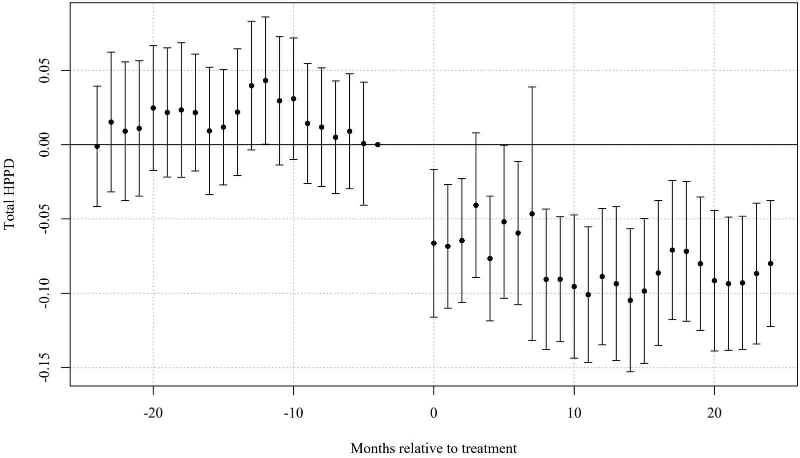
Main Results: Event Study (LPN HPPD)



Main Results: Event Study (CNA HPPD)

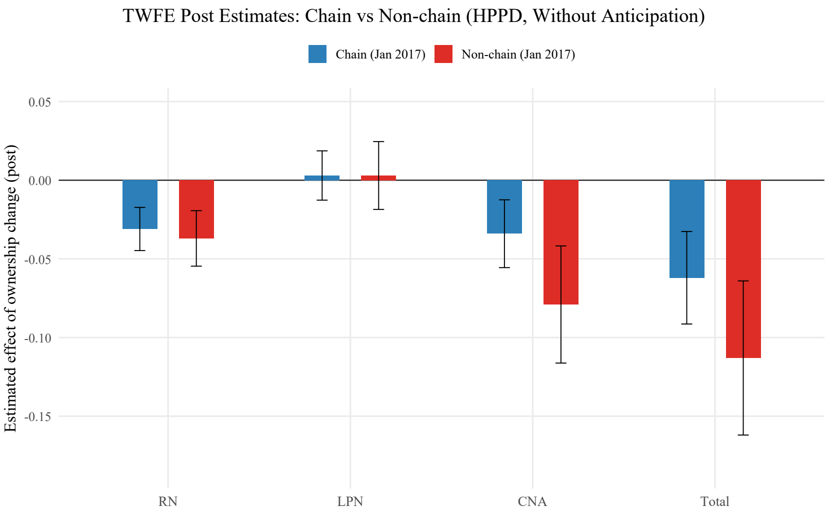


Main Results: Event Study (Total Nursing HPPD)

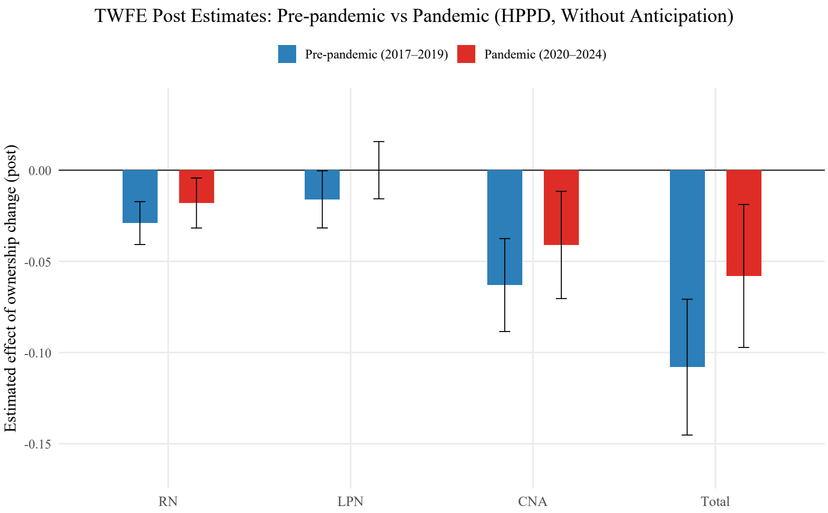


Average Post-Change Staffing (TWFE DiD)

Heterogeneity: Chain vs Non-Chain Facilities



Heterogeneity: Pre vs Pos-Pandemic



Robustness: Anticipation and the Donut Window

Additional Robustness Checks (Summary)